

Cardiovascular Risk Reduction

Dr. Muhammad Reihan, Internal Medicine | Source: Reihan_CVD_RR.pdf (39 slides)

Legend: ★ = high-yield / likely exam target · [added] = not on slides. Everything else comes from the lecture.

Slides 21 ("If You Quit Smoking Right Now") and 24 (HTN questions) have **no answers on the slide**: slide 21's signposts are blank and slide 24's questions are unanswered, so the content was probably delivered verbally. Those gaps are flagged below.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Classify modifiable vs non-modifiable ASCVD risk factors	"Which is non-modifiable?"
Knowledge	★ 10-year ASCVD risk categories and how they guide therapy	Cut-offs 5 / 7.5 / 20%
Skill	Calculate 10-year ASCVD risk → statin & aspirin decisions	Case vignette (as in the lecture's closing case)
Skill	Individualized plan: lifestyle + pharmacotherapy	Statin eligibility, aspirin USPSTF grades

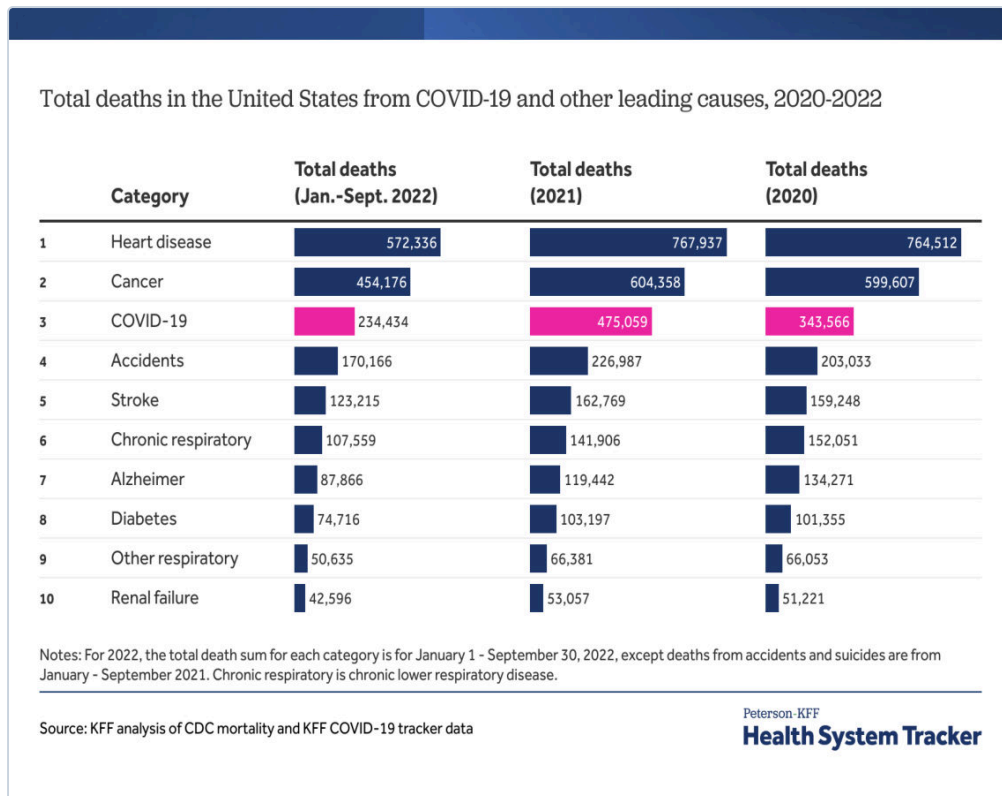
Lecture flow: Overview/epidemiology → risk factors → 10-y ASCVD categories → Lifestyle (diet → physical activity → tobacco) → HTN → Hyperlipidemia → DM → Aspirin → Summary → Case.

2. Main Content (Lecture Order)

2.1 Overview & Epidemiology (Slides 5-8)

Critical-thinking prompt: how many relatives have had a heart attack or stroke, and how did it affect their lives?

- **CHD → > 2.5 million hospitalizations/year** worldwide
- US: an **MI every 34 seconds; one death each minute** from a major coronary event
- ★ CVD = **leading cause of mortality globally**. It includes **HTN, CAD, HF, congenital defects, strokes**.



Slide 7: Leading causes of death, US 2020-2022 (heart disease #1)

★ Why MI death rates have declined:

1. Heightened public awareness
2. ↑ Availability of **AEDs**
3. Improved CV diagnosis & therapy
4. CV drugs in high-risk persons
5. Improved revascularization
6. More aggressive risk-factor modification

🔄 **Recap: Overview.** CVD is the #1 killer (MI every 34 s in the US). Deaths are falling because of awareness, AEDs, better diagnosis, drugs, revascularization, and risk-factor control.

🔗 **Link:** The last reason on that list, risk-factor modification, is the subject of the whole lecture. To modify risk, first identify it.

▶ **Up next: Traditional ASCVD risk factors.** Which can be changed and which cannot.

2.2 Traditional ASCVD Risk Factors (Slide 9)

Non-modifiable	Modifiable
Advancing age	Smoking
Male sex	Diabetes mellitus
Race and ethnicity	Hypertension
	Dyslipidemia (↑ total cholesterol, ↓ HDL)

🔄 **Recap: Risk factors.** Three non-modifiable (age, male sex, race/ethnicity) and four modifiable (smoking, DM, HTN, dyslipidemia).

🔗 **Link:** Individual factors don't tell you how much risk a person carries. They are combined into a single 10-year risk estimate that decides treatment.

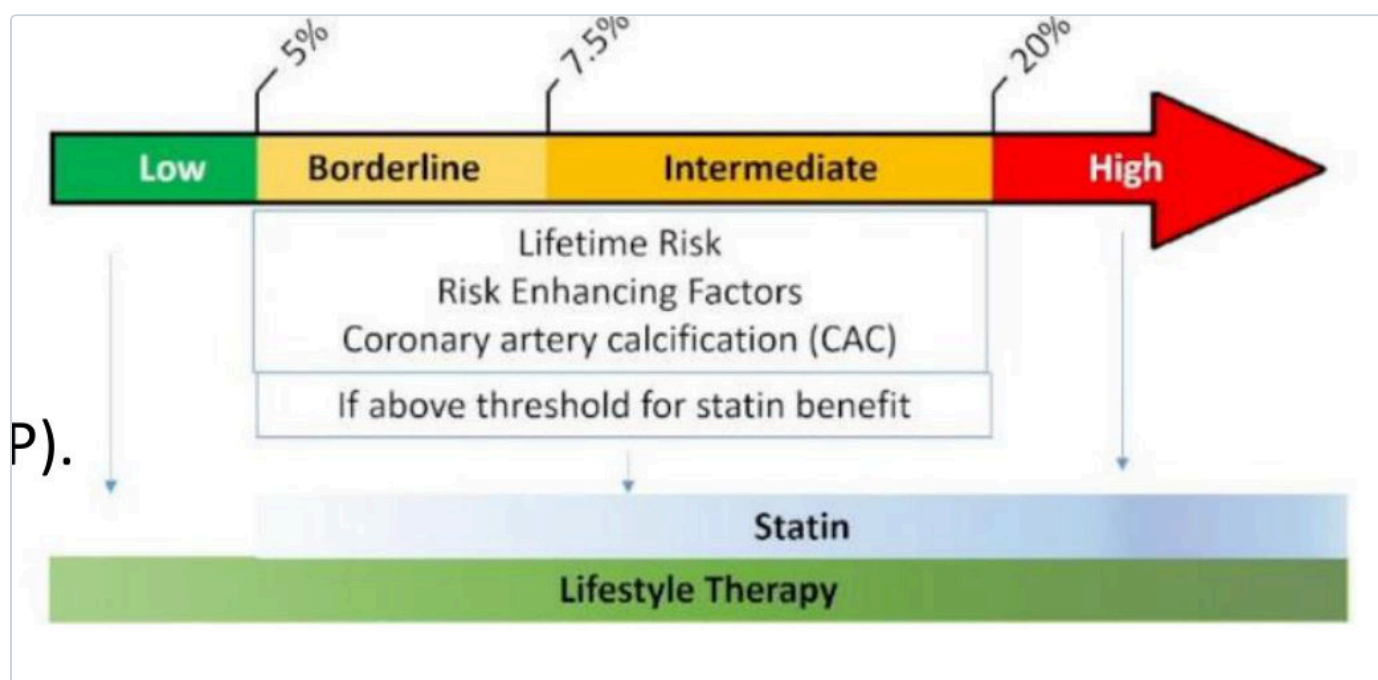
👉 **Up next: 10-year ASCVD risk categories.**

2.3 10-Year ASCVD Risk Categories (Slide 10)

Estimates the risk of **MI or stroke over 10 years**. Used for ages ★ **40-75**. Calculated with the AHA/ACC 2013 pooled cohort calculator (MDCalc).

Category	10-y risk
Low	< 5%
Borderline	5-7.4%
Intermediate	7.5-19.9%
High	≥ 20%

Mnemonic [added]: "**5 - 7.5 - 20**", the three cut-offs.



Slide 29: Risk bands → lifestyle for all, statin above threshold

🔄 **Recap: 10-y ASCVD.** Ages 40-75. Low <5, borderline 5-7.4, intermediate 7.5-19.9, high ≥20%.

🔗 **Link:** Whatever the risk category, **lifestyle is the foundation for everyone**. Drugs are added on top based on risk. The lecture therefore starts treatment with lifestyle, beginning with diet.

👉 **Up next: Diet.** Specific intake targets, the Mediterranean diet, and tailored counseling.

2.4 Lifestyle: Life's Simple 7 & Diet (Slides 11-15)

★★ **Orientation: AHA Life's Simple 7 (Slide 11)**

#	Goal	Ideal target (slide)
1	Get active	≥ 150 min/week moderate or ≥ 75 min/week vigorous activity
2	Eat better	Healthy diet: 4-5 components of the healthy diet score*
3	Lose weight	BMI < 25
4	Stop smoking	Never smoked or quit > 1 year ago
5	Control cholesterol	Total cholesterol < 200 mg/dL
6	Manage blood pressure	BP < 120/80 mm Hg
7	Reduce blood sugar	Fasting glucose < 100 mg/dL

Healthy diet score components: (1) ≥ 4.5 cups/day fruit & vegetables; (2) ≥ 2 servings (3.5 oz) of fish/week; (3) ≥ 3 servings/day whole grains; (4) sodium < 1500 mg/day; (5) ≤ 36 oz/week sugar-sweetened beverages* (Lloyd-Jones, Circulation 2010).

Mnemonic [added]: the first 4 are **behaviours** (active, eat, weight, smoking) and the last 3 are **numbers** (cholesterol 200, BP 120/80, glucose 100). ⚠ Don't mix up the **ideal** BP in Life's Simple 7 (< 120/80) with the **treatment target** (< 130/80).

AHA Life's Simple 7



Life's Simple 7

1. Get Active
2. Eat Better
3. Lose Weight
4. Stop Smoking
5. Control Cholesterol
6. Manage Blood Pressure
7. Reduce Blood Sugar

1. ≥150 minutes moderate activity /week or ≥75 minutes vigorous activity/week
2. Eat a healthy diet (4-5 components of healthy diet score*)
2. Have a normal body weight (BMI < 25)
3. Never smoked or quit >1 year ago
4. Total cholesterol <200 mg/dL
5. Blood pressure <120/80 mm Hg
6. Fasting blood glucose <100 mg/dL

* 1) 4.5 cups or more of fruits and vegetables per day 2) two or more 3.5-oz servings of fish per week 3) three servings per day of whole grains 4) less than 1500 mg of sodium per day 5) 36 ounces or less of sugar-sweetened beverages per week

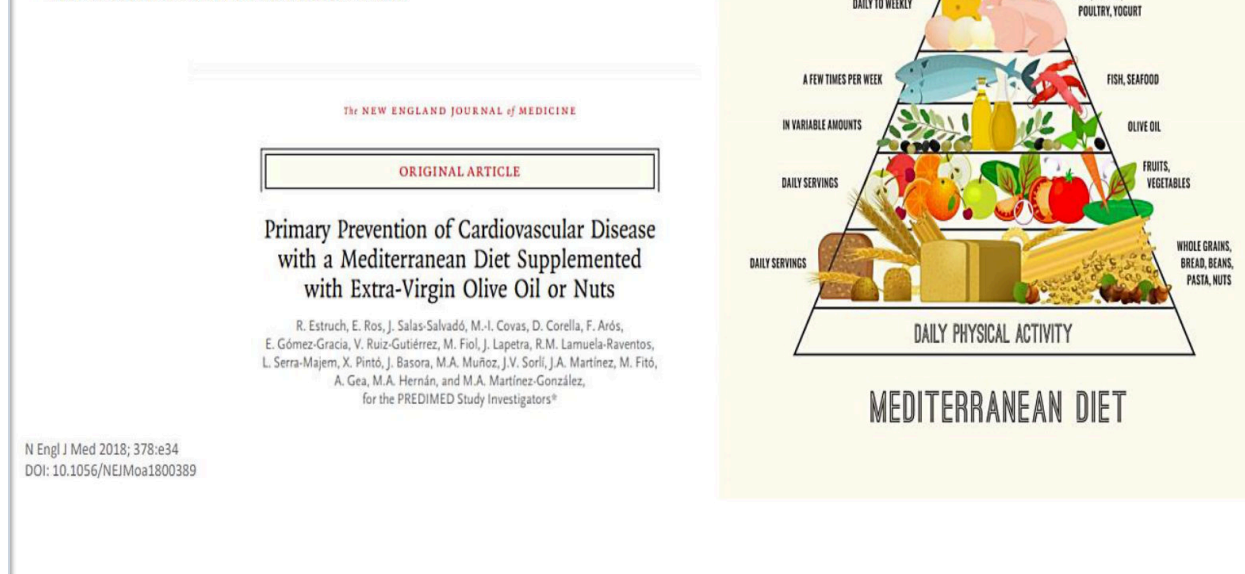
Lloyd-Jones et al. Circulation 2010; 121:596-813

Slide 11: AHA Life's Simple 7

★ Recommended intakes (Slide 12):

Component	Target
Saturated fat	< 10% of total energy, replaced by polyunsaturated fat
Trans fat	< 1% of total energy
Salt	< 5 g/day
Fibre	30-45 g/day (wholegrain, fruit, vegetables)
Fruit	200 g/day (2-3 servings)
Vegetables	200 g/day (2-3 servings)
Fish	≥ 2×/week, one oily

- Numerous observational studies have demonstrated the benefit of a Mediterranean style diet on cardiovascular risk reduction



Slide 13: Mediterranean diet and the PREDIMED trial

- ★ **Mediterranean diet**: observational benefit for CV risk reduction. **PREDIMED** (NEJM 2018): primary prevention with a Mediterranean diet supplemented with **extra-virgin olive oil or nuts**.
- Nutritional counseling: food pyramid (base = daily exercise, portion & weight control, then vegetables/fruit/whole grains/healthy oils; top = limit **red meat, butter, sugary drinks, sweets, salt**).

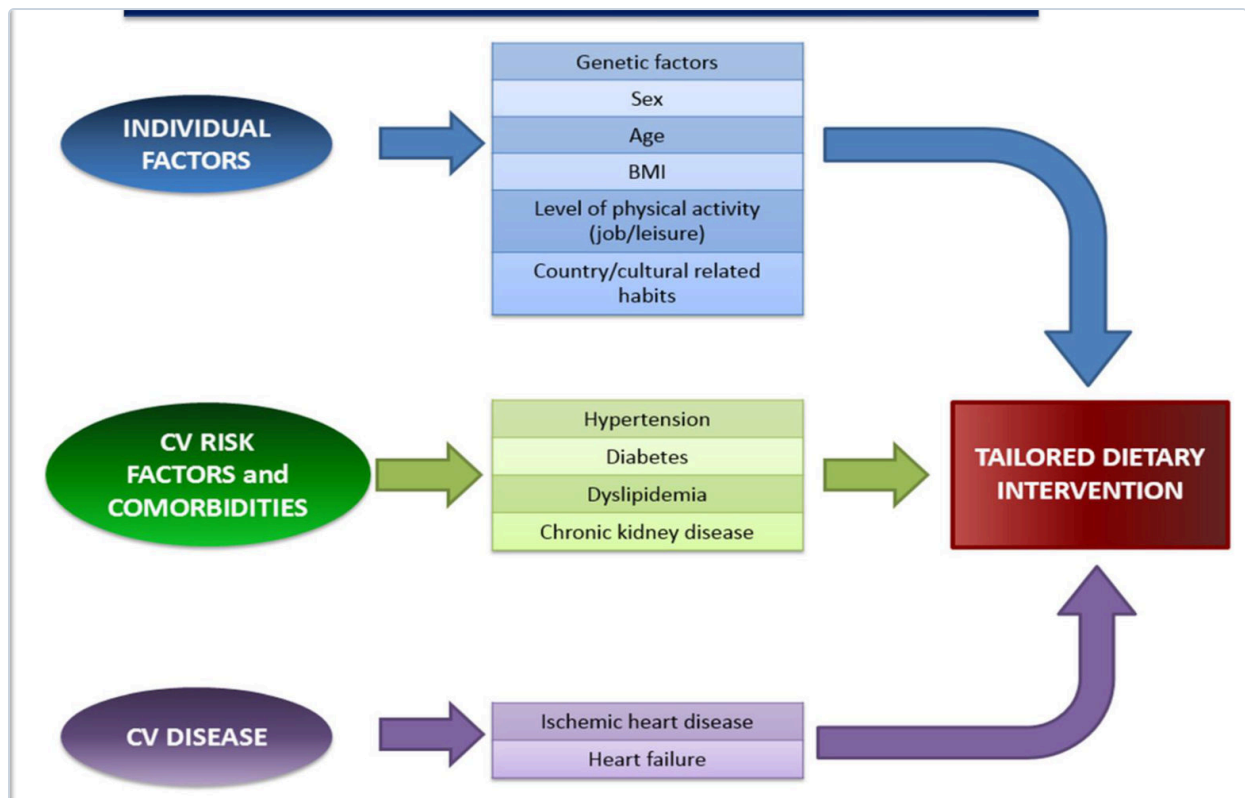


Slide 14: Nutritional counseling: healthy pyramid vs junk food

- ★ **Tailored dietary intervention** combines 3 inputs:
 - Individual factors**: genetics, sex, age, BMI, physical activity (job/leisure), cultural habits

2. **CV risk factors & comorbidities:** HTN, DM, dyslipidemia, CKD

3. **CV disease:** IHD, HF



Slide 15: Tailored dietary intervention: three inputs

Mnemonic [added]: the "<10, <1, <5" rule: sat fat <10%, trans fat <1%, salt <5 g.

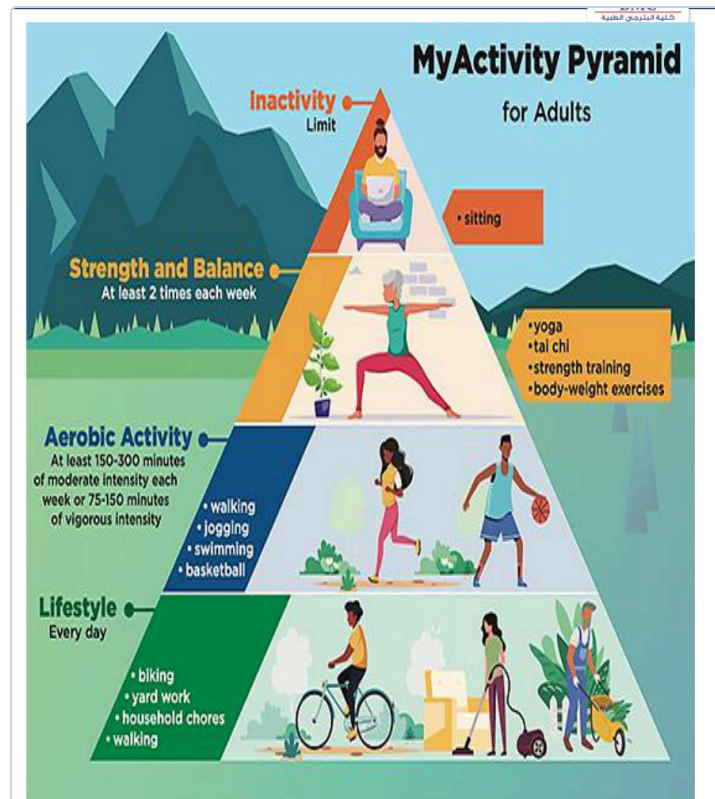
🔄 **Recap: Diet.** <10% saturated, <1% trans, <5 g salt, 30–45 g fibre, 200 g fruit + 200 g veg, fish twice weekly (one oily). Mediterranean diet (PREDIMED). Tailor to the individual, their risk factors, and their disease.

🔗 **Link:** Diet is energy in. Physical activity is energy out, and the second lifestyle pillar.

▶ **Up next: Physical activity.** Prescription, session structure, Saudi data.

2.5 Lifestyle: Physical Activity (Slides 16-19)

- Aerobic exercise should be offered to patients with **known CAD**
- ★ **Stable angina or stable chronic HF:** **moderate-to-vigorous** aerobic training $\geq 3 \times/\text{week}$, **30 min/session**
- **Sedentary** patients: strongly encourage starting **light-intensity** programs



Slide 16: Physical activity pyramid for adults

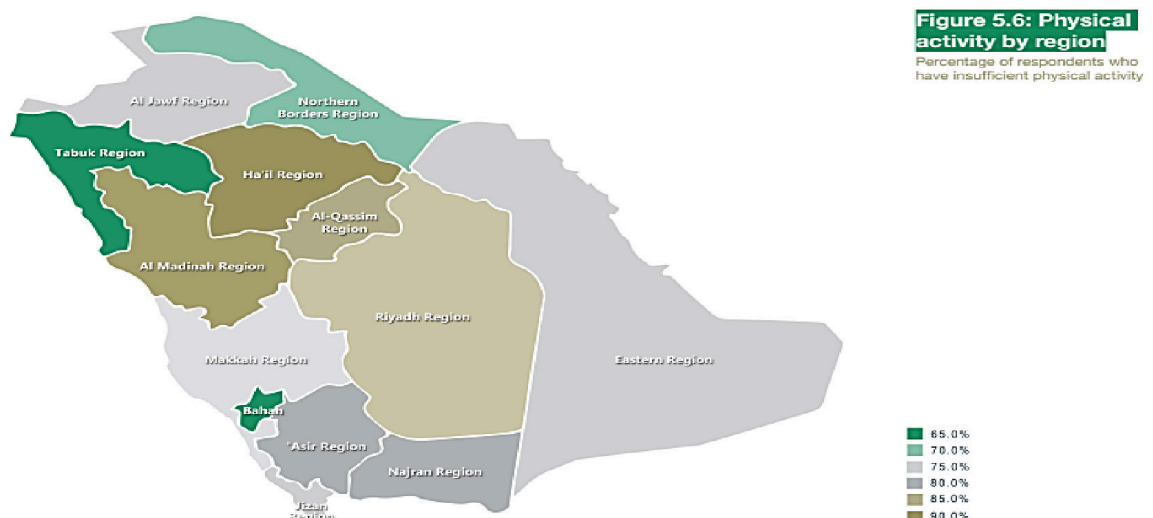
★ Typical exercise session:

1. **Warm-up: 5-10 min**
2. **Aerobic conditioning: up to 30 min or more** (walking, jogging, cycling, stairs)
3. **Cool-down: 5-7 min**

KSA physical-activity insufficiency (MOH World Health Survey):

Highest insufficiency	Lowest insufficiency
Hail 91%, Madinah 90%, Qasim 88%	Tabuk 64%, Bahah 65%

- A large variation in physical activity insufficiency exists between regions. Hail (91%), Madinah (90%), and Qasim (88%) have the highest percentages, while Tabuk (64%) and Bahah (65%) have the lowest.



<https://www.moh.gov.sa/en/Ministry/Statistics/Population-Health-Indicators/Documents/World-Health-Survey-Saudi-Arabia.pdf>

Slide 18: Physical activity insufficiency by Saudi region

★ Take-home:

- Lifestyle improvement should **begin early** and underlie **all** preventive efforts
- **Most CV events are preventable** with lifestyle modification
- ★ **"Some is better than none; more is even better"**

🔄 **Recap: Physical activity.** $\geq 3 \times / \text{week} \times 30 \text{ min}$ moderate-vigorous for stable CAD/HF. Session = warm-up 5-10, aerobic ~30, cool-down 5-7. KSA inactivity is highest in Hail (91%). Some beats none.

🔗 **Link:** Diet and exercise are the things to add. The next lifestyle change is removing something, and it is the single most powerful one: smoking.

▶ **Up next: Tobacco cessation.** Mortality benefit, time course, and pharmacotherapy safety in CAD.

2.6 Lifestyle: Tobacco Cessation (Slides 20-22)

- ★★ **Smoking cessation = the MOST important and MOST cost-effective** lifestyle modification for preventing CVD
- In patients with prior **MI, CABG, angioplasty, or known CHD**, quitting reduces mortality: **RR 0.64 (CI 0.58-0.71)**, i.e., ★ **36% ↓ mortality after MI**
- ★ **Time course:** mortality risk **↓ 50% in the first couple of years**. It **approaches non-smokers' risk in ~5-15 years** (Critchley & Capewell, 2012).
- Eliminating health-risk behaviours could prevent **$\geq 80\%$ of CVD** and **40% of cancers** (BACPR, 2017)
- (Slide 21 "If You Quit Smoking Right Now" has blank signposts in the PDF, so its content was not captured.)

★ Pharmacotherapy in CAD:

Agent	Safety in CAD
Nicotine replacement therapy	★ Safe , should be routinely offered
Bupropion	Safe in stable CAD (some studies)
Varenicline	Safe in some studies, but a meta-analysis found a small, statistically significant ↑ CVD

🔄 **Recap: Smoking.** Most important and most cost-effective. RR 0.64 = 36% ↓ mortality post-MI. Risk halves in ~2 years and approaches a non-smoker's in 5-15 years. NRT is safe. Varenicline has a CVD signal.

🔗 **Link:** That completes lifestyle. The lecture now turns to the modifiable disease risk factors that need targets and often drugs, starting with the #1 CV risk factor worldwide.

▶ **Up next: Hypertension.**

2.7 Hypertension (Slides 23-24)

- ★★ **HTN = the number one cardiovascular risk factor** and the **world's greatest risk factor for death and disability**
- KSA prevalence: **mid-20% range**

- ★ People with HTN are **3× more likely to die of heart disease** and **4× more likely to die of stroke**
- ★ Target (Summary slide 35): **BP < 130/80 mm Hg for most patients**

Discussion questions posed (Slide 24), unanswered on the slide:

1. Is **home monitoring** superior to in-office monitoring?
2. What is the **optimal BP target?** (Answered on slide 35: **< 130/80**)
3. Do different **drug classes** have better outcomes?
4. Is it better to start with **one drug or more than one class?**

[added, not on slides; verify against your course: home/ambulatory BP is better than office BP for diagnosis and detects white-coat and masked HTN. BP reduction itself matters more than which first-line class (thiazide, ACEi/ARB, CCB) is used. Starting with 2 drugs is reasonable when BP is well above target (e.g., stage 2).]

🔄 **Recap: HTN.** #1 CV risk factor. Mid-20% prevalence in KSA. 3× cardiac death, 4× stroke death. Target <130/80.

🔗 **Link:** HTN damages the vessel wall. Lipids supply the material that builds the plaque, making hyperlipidemia the second major modifiable driver of atherosclerosis.

👉 **Up next: Hyperlipidemia.** Diagnosis cut-offs, risk factors, treatment, and exactly who gets a statin.

2.8 Hyperlipidemia (Slides 25-30)

- ★ One of the **most important (and modifiable)** CAD risk factors → **accelerated atherosclerosis**
- KSA: **43%** of survey respondents have **raised serum cholesterol**

Risk factors (Slide 26): diet, age, DM, inactive lifestyle, family history, gender, medications, genetic mutations

★ Diagnosis (Slide 27):

- **Fasting or non-fasting** lipid profile (TC, HDL, TG)
- ★ Pathological values on **2 separate occasions** are needed to confirm
- ★ **Dyslipidemia = LDL > 130 mg/dL and/or HDL < 40 mg/dL**

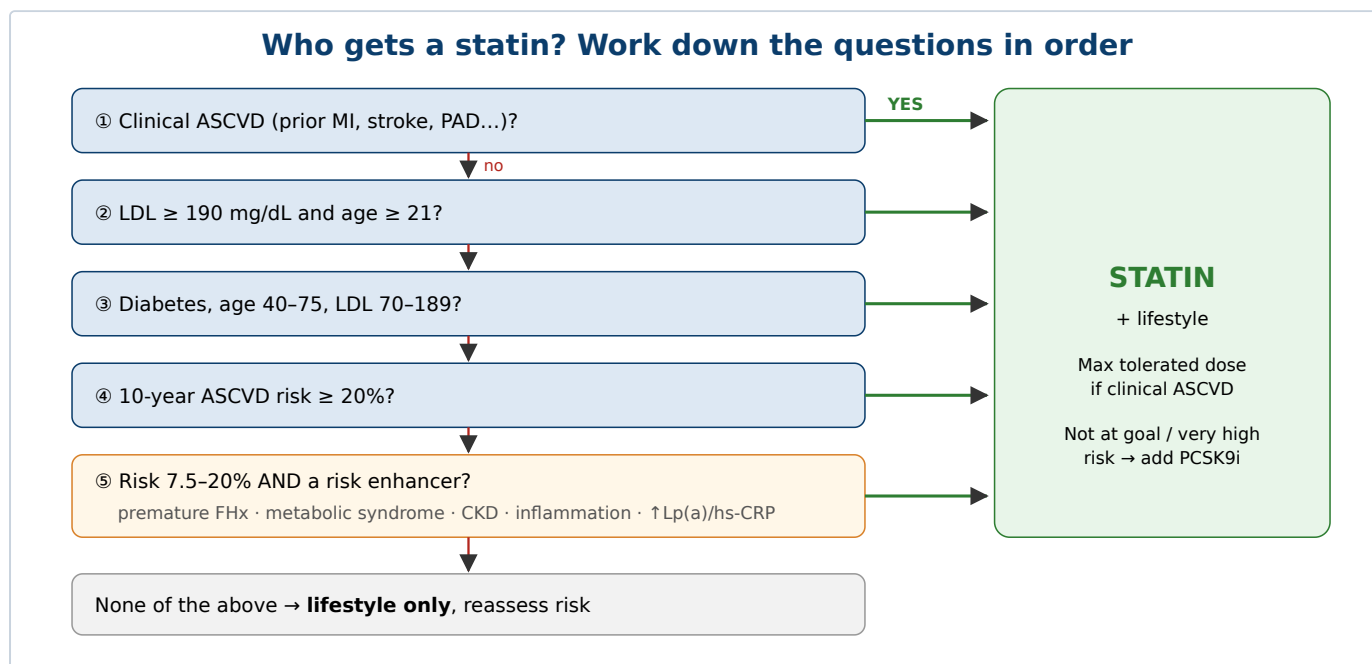
Parameter	Optimal	Pathological
Total cholesterol	200 mg/dL (i.e., < 200)	Borderline 200-239 ; High > 240
TG	150 mg/dL (i.e., < 150)	Borderline 150-199 ; High > 200 ; Very high ≥ 500
LDL	100 mg/dL (i.e., < 100)	Near-optimal 100-129 ; Borderline high > 130 ; High > 160 ; Very high ≥ 190
HDL	≥ 60 mg/dL	Low < 40
LDL/HDL ratio	♂ < 3.5 ; ♀ < 3.0	♂ > 3.5 ; ♀ > 3.0

Management (Slide 28):

- Non-pharmacological: dietary changes, **↓ saturated fat & cholesterol**, **eliminate alcohol**, healthy weight, physical activity
- Medical: ★ **Statins** (first line) → **PCSK9 inhibitors** (second line if poor statin response)

★★★★ **Who gets a statin? (Slide 29).** The **4 statin benefit groups** [the term "benefit groups" is added]:

#	Group
1	Clinical ASCVD
2	LDL $\geq$ 190 mg/dL, age $\geq$ 21
3	Diabetics, LDL 70-189, age 40-75
4	10-y ASCVD risk $\geq$ 20%
(+)	ASCVD 7.5-20% + risk enhancers: family history of premature ASCVD · metabolic syndrome · CKD · inflammatory conditions · elevated biomarkers (lipoproteins, hs-CRP)



Added diagram: Statin decision ladder

Mnemonic [added]: "**A-190-DM-20**": **ASCVD**, LDL **190**, **DM** 40-75, risk **20**%.

★ Take-home (Slide 30):

- **First assess ASCVD risk**
- Statins + lifestyle in higher-risk patients; **maximally tolerated dose** for the highest risk (clinical ASCVD)
- **PCSK9i reduce events when added to statins** in **very-high-risk secondary-prevention** patients
- ★ "**Lower is better**"

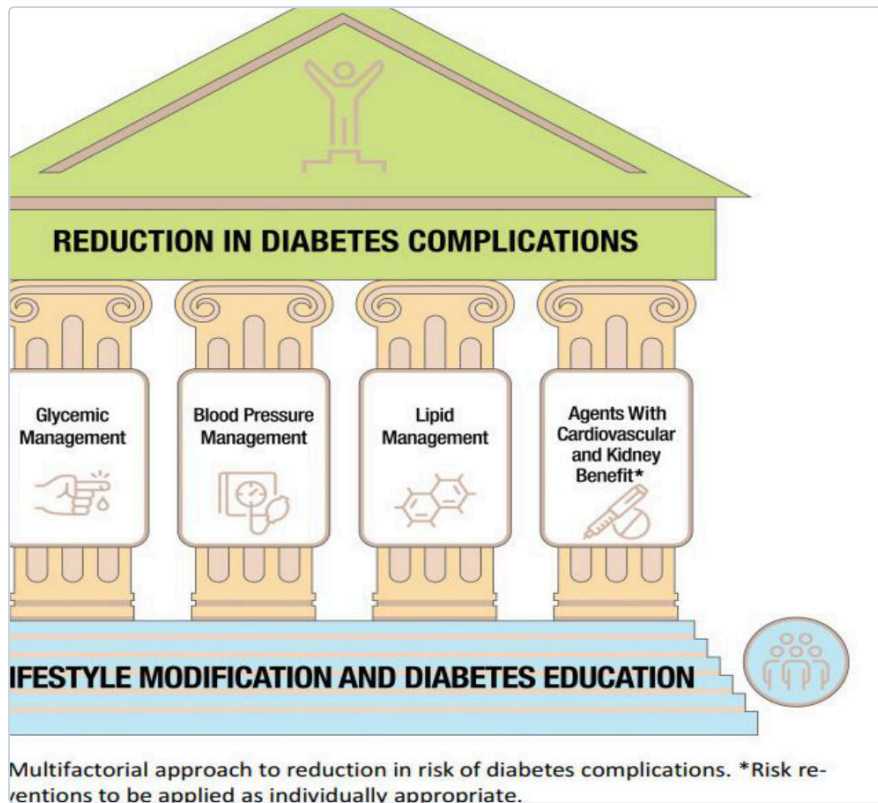
🔄 **Recap: Hyperlipidemia.** Diagnose on 2 occasions (LDL >130 and/or HDL <40). Statin groups: ASCVD, LDL $\geq$ 190, DM 40-75 with LDL 70-189, risk $\geq$ 20% (or 7.5-20% + enhancers). Statin first, then PCSK9i. Lower is better.

🔗 **Link:** Diabetes appeared inside the statin criteria, and it is itself a major CV risk multiplier. The lecture now asks how treating DM itself can protect the heart.

👉 **Up next: Diabetes mellitus.** CV risk, and the cardioprotective drug classes (SGLT-2i, GLP-1 RA).

2.9 Diabetes Mellitus (Slides 31-33)

- ★ DM → **2-3× ↑ CVD risk**
- CVD is a more likely cause of hospitalization** in diabetics than diabetes-related causes
- ★ Experts recommend **prevention of CV events as the goal** of diabetes treatment
- ★ **SGLT-2 inhibitors** and **GLP-1 receptor agonists** showed **CV protection** in outcome trials (T2DM with high CV risk)
- ★ Consider them when **ASCVD, HF, or CKD predominates, independent of HbA1c**
- Dapagliflozin** (SGLT-2i) is added if **metformin monotherapy** fails to reach **HbA1c < 7%**. In the lecture's case (HbA1c **6.2%** on metformin, no ASCVD/HF/CKD) it is **not needed**.
- ADA: **lifestyle management + medications**, including nutrition therapy, physical activity, smoking cessation



Slide 33: Reducing diabetes complications: multifactorial approach

🔄 **Recap: DM.** 2-3× CVD risk. The goal of treatment is preventing CV events. SGLT-2i/GLP-1 RA for ASCVD, HF, or CKD regardless of HbA1c. Otherwise add them only if HbA1c is $\geq 7\%$ on metformin.

🔗 **Link:** The last preventive tool is the most commonly misused: aspirin. Unlike statins, its benefit is small and it carries a bleeding risk, so the rules are restrictive.

👉 **Up next: Aspirin for primary prevention (USPSTF 2023).**

2.10 Aspirin for Primary Prevention (Slide 34, USPSTF 2023)

Population	Recommendation	Grade
Adults 40-59 with $\geq 10\%$ 10-y CVD risk	Individual decision. Net benefit small . More likely to benefit if not at ↑ bleeding risk and willing to take it daily.	★ C
Adults ≥ 60	Recommend AGAINST initiating low-dose aspirin for primary prevention	★ D

The U.S. Preventative Services Task Force (USPSTF) 2023

Population	Recommendation	Grade
Adults aged 40 to 59 years with a 10% or greater 10-year cardiovascular disease (CVD) risk	The decision to initiate low-dose aspirin use for the primary prevention of CVD in adults aged 40 to 59 years who have a 10% or greater 10-year CVD risk should be an individual one. Evidence indicates that the net benefit of aspirin use in this group is small. Persons who are not at increased risk for bleeding and are willing to take low-dose aspirin daily are more likely to benefit.	C
Adults 60 years or older	The USPSTF recommends against initiating low-dose aspirin use for the primary prevention of CVD in adults 60 years or older.	D

Slide 34: USPSTF 2023 aspirin recommendation

Mnemonic [added]: "**40-59 & 10 = maybe (C); 60+ = don't (D).**"

Recap: Aspirin. Primary prevention: 40–59 with $\geq 10\%$ risk is an individual choice (C). Age ≥ 60 : don't start (D). Avoid in low-risk patients and the elderly.

Link: The lecture now pulls everything into a summary and tests it with a case that combines ASCVD risk, DM, lipids, and aspirin.

Up next: Summary + case.

2.11 Summary (Slide 35)

- CVD is the **leading cause of death worldwide**
- Assess every patient's CV risk**
- ★ **Lifestyle = the most important component** of prevention and risk-factor control ("**Life's Simple 7**")
 - See §2.4 for the 7 goals and targets (Slide 11). [added: AHA updated it to "**Life's Essential 8**" (adds sleep) in 2022.]
- Statins** + lifestyle in higher-risk patients; **PCSK9i** for the **very highest** risk
- ★ **BP target < 130/80** for most
- ★ **Low-dose aspirin only in higher-risk** patients **if benefit > bleeding risk. Avoid in low-risk patients and the elderly.**

3. High-Yield Comparison Tables

Table A: Risk factors

Non-modifiable	Modifiable (traditional)	Modifiable (lifestyle)
Age, male sex, race/ethnicity	Smoking, DM, HTN, dyslipidemia	Diet, physical inactivity, alcohol, weight

Table B: 10-y ASCVD risk → action (lecture content combined)

Risk	Statin	Aspirin (primary prevention)
< 5% (low)	Lifestyle only (unless DM/LDL ≥190/ASCVD)	Avoid
5-7.4% (borderline)	Lifestyle (unless another statin group applies)	Avoid
7.5-19.9% (intermediate)	Statin if risk enhancers	≥10% and age 40-59 → individual (C)
≥ 20% (high)	★ Statin	≥10% and age 40-59 → individual (C)
Any risk, age ≥ 60	Per criteria	★ Against (D)

Table C: Lipid cut-offs (see §2.8)

Table D: Statin vs PCSK9i vs aspirin vs SGLT-2i/GLP-1 RA

Drug	Who (per lecture)	Key point
Statin	4 benefit groups + intermediate risk with enhancers	First line; max tolerated dose if clinical ASCVD
PCSK9i	Poor statin response / very-high-risk secondary prevention	Add-on to statin
Aspirin	40-59, ≥10% risk, low bleeding risk	C / D grades
SGLT-2i / GLP-1 RA	T2DM with ASCVD, HF, or CKD	Independent of HbA1c

Table E: Smoking-cessation timeline

Time after quitting	Effect
First couple of years	Mortality risk ↓ 50%
5-15 years	Approaches non-smoker risk
Post-MI (overall)	RR 0.64 → 36% ↓ mortality

Table F: Lifestyle prescriptions

Domain	Numbers
Diet	Sat fat <10%, trans <1%, salt <5 g, fibre 30-45 g, fruit 200 g, veg 200 g, fish ≥2×/wk
Exercise	≥3×/wk × 30 min moderate-vigorous; warm-up 5-10, aerobic ~30, cool-down 5-7
BP	<130/80

Table G: KSA statistics

Metric	Value
HTN prevalence	Mid-20% range
Raised cholesterol	43%
Physical-activity insufficiency, highest	Hail 91%, Madinah 90%, Qasim 88%
Physical-activity insufficiency, lowest	Tabuk 64%, Bahah 65%

4. Rules, Exceptions, Patterns

- **Lifestyle for everyone, first and always.** Drugs are added according to risk.
- **The 10-y ASCVD calculator is for ages 40-75.** Exceptions: **LDL ≥ 190 (age ≥ 21)** and **clinical ASCVD** get statins **without** calculating risk.
- **DM aged 40-75 with LDL 70-189 $\rightarrow$ statin**, even if 10-y risk is low.
- **Intermediate risk (7.5-20%) $\rightarrow$ statin only with risk enhancers.**
- **Dyslipidemia needs 2 abnormal measurements**, fasting or non-fasting.
- **Aspirin for primary prevention is the exception, not the rule.** Consider only at 40-59 with $\geq 10\%$ risk and low bleeding risk. **Never start at ≥ 60 .**
- **SGLT-2i/GLP-1 RA:** the indication in ASCVD, HF, or CKD is **independent of HbA1c**. Otherwise escalate only if HbA1c $\geq 7\%$ on metformin.
- **Smoking cessation is the #1 lifestyle measure; HTN is the #1 risk factor.** Don't mix them up.
- **NRT is safe in CAD.** Varenicline has a small CVD signal (meta-analysis).
- **Lipids: "lower is better."**

5. Clinical Correlations

- **Post-MI smoker:** quitting gives a 36% mortality reduction, larger than many drugs [*comparison added*]. Offer NRT routinely.
- **Diabetic with established ASCVD/HF/CKD:** add SGLT-2i/GLP-1 RA even if HbA1c is at target.
- **Patient with LDL ≥ 190** (e.g., familial hypercholesterolemia [*added*]): high-intensity statin from age 21, no calculator needed.
- **Elderly patient asking to start aspirin "for the heart":** USPSTF D, do not start (bleeding outweighs benefit). [*added: this concerns primary prevention only. Aspirin in established CAD (secondary prevention) remains indicated, per the CAD lecture.*]
- **Stable CAD/HF patient:** prescribe structured aerobic exercise ($\geq 3\times/\text{wk}$, 30 min) as cardiac rehabilitation.
- **KSA context:** high inactivity ($\geq 88\%$ in Hail/Madinah/Qasim) and 43% raised cholesterol make population-level lifestyle measures the priority.

6. Rapid Revision (Last-Day)

- CVD is the #1 global killer. US: an MI every **34 s**. CHD: **>2.5 M** hospitalizations/year.
- Non-modifiable: **age, male sex, race**. Modifiable: **smoking, DM, HTN, dyslipidemia**.
- 10-y ASCVD (age **40-75**): **<5 low $\cdot$ 5-7.4 borderline $\cdot$ 7.5-19.9 intermediate $\cdot$ ≥ 20 high.**
- Diet: **sat $<10\%$, trans $<1\%$, salt <5 g, fibre 30-45 g, fruit 200 g, veg 200 g, fish $\geq 2\times/\text{wk}$ (1 oily).** Mediterranean diet (**PREDIMED**).
- Exercise: **$\geq 3\times/\text{wk} \times 30$ min.** Warm-up **5-10**, cool-down **5-7**. "Some $>$ none; more $>$ some."
- Smoking cessation = **most important and cost-effective. RR 0.64 (36% $\downarrow$).** Risk halves in ~ 2 years and equals a non-smoker's in **5-15 y. NRT safe.**
- Eliminating risk behaviours prevents **$\geq 80\%$ of CVD** and **40% of cancers**.
- HTN = **#1 CV risk factor. 3 $\times$ heart death, 4 $\times$ stroke death.** Target **$<130/80$** . KSA **mid-20%**.
- Lipids: dyslipidemia = **LDL >130 and/or HDL <40** , confirmed on **2 occasions**. KSA **43%** raised cholesterol.
- Statin groups: **ASCVD $\cdot$ LDL ≥ 190 (≥ 21 y) $\cdot$ DM 40-75 + LDL 70-189 $\cdot$ risk $\geq 20\%$ $\cdot$ 7.5-20% + enhancers.**

- PCSK9i = add-on for very-high-risk patients. **Lower is better.**
- DM = **2-3×** CVD. **SGLT-2i/GLP-1 RA** if ASCVD/HF/CKD, **regardless of HbA1c.**
- Aspirin (USPSTF 2023): **40-59 & ≥10% → C (individual); ≥60 → D (against).**
- **Life's Simple 7:** active ≥150 min/wk · healthy diet · BMI <25 · no smoking (>1 y) · TC <200 · BP <120/80 · FBG <100.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slides 36-37). 54 ♂, T2DM on metformin, **HbA1c 6.2%**, diverticulosis, maternal aunt had an MI at 63, non-smoker, 3-4 beers/week, BMI 24, **BP 124/68**. TC 185, **HDL 45, LDL 122**, TG 140. **10-y risk 9.1%**. Besides lifestyle counseling, the most appropriate recommendation? A) Aspirin **B) Atorvastatin** C) Dapagliflozin D) No further action E) PCSK9i

Answer: B. He is **diabetic, age 40-75, LDL 70-189 (122)** → statin benefit group regardless of risk %.

- **A X:** 10-y risk **9.1% < 10%**, so he doesn't meet the USPSTF threshold.
- **C X:** HbA1c **6.2% < 7%** on metformin with **no ASCVD, HF, or CKD.**
- **D X:** he meets a statin indication.
- **E X:** PCSK9i only for poor statin response or very-high-risk secondary prevention.
- Distractor: the **aunt is not a first-degree relative** (and 63 is not "premature" for a female), so this is not a family-history risk enhancer *[reasoning added]*.

Q2 (MCQ). Which lifestyle intervention is described as the **most important and most cost-effective** for preventing CVD? a) Mediterranean diet b) Aerobic exercise ≥3×/week **c) Smoking cessation** d) Salt < 5 g/day **Answer: c.** Stated explicitly (Slide 20). Don't confuse it with "HTN = #1 risk factor".

Q3 (MCQ). A 65-year-old woman with a 10-y ASCVD risk of 12% and no bleeding history asks whether to start low-dose aspirin for primary prevention. Best advice? **Answer: Recommend against (USPSTF D, age ≥60).** The ≥10% risk criterion applies only to ages **40-59** (grade C).

Q4 (Short answer). List the statin indications taught in the lecture. **Answer:** (1) Clinical ASCVD. (2) LDL ≥190 mg/dL, age ≥21. (3) Diabetics aged 40-75 with LDL 70-189. (4) 10-y ASCVD risk ≥20%. (5) Risk 7.5-20% plus risk enhancers: family history of premature ASCVD, metabolic syndrome, CKD, inflammatory conditions, elevated biomarkers (lipoproteins, hs-CRP).

Q5 (MCQ). A patient's lipid profile shows LDL 145 mg/dL and HDL 50 mg/dL on one measurement. Next step? **Answer: Repeat the lipid profile on a second occasion** before diagnosing dyslipidemia (2 pathological values are required). LDL 145 is "borderline high" (>130).

8. Exam Pitfalls & Traps

1. **"#1 risk factor" (HTN) vs "most important/cost-effective intervention" (smoking cessation).** These are different questions.
2. **Aspirin for everyone at risk.** No: 40-59 with ≥10% risk only (C). **≥60 is D.** Low-risk patients: avoid.

3. **Applying the 10-y calculator outside ages 40-75**, or to patients with LDL ≥ 190 or ASCVD (they get statins regardless).
4. **Forgetting the DM statin rule.** A diabetic aged 40-75 with LDL 70-189 needs a statin even if the 10-y risk is under 7.5%.
5. **Intermediate risk $\neq$ automatic statin.** Risk enhancers are required.
6. **Adding an SGLT-2i because the patient "has diabetes."** It is needed only with ASCVD/HF/CKD or HbA1c $\geq 7\%$ on metformin.
7. **Diagnosing dyslipidemia from a single value.** Two occasions are required. Fasting is **not** mandatory.
8. **Premature family history** means a **first-degree** relative ($\sigma < 55$, $\phi < 65$, per the CAD lecture). Aunts, uncles and grandparents don't count.
9. **Varenicline "totally safe" in CAD.** The meta-analysis showed a small $\uparrow$ CVD. **NRT is the safe routine choice.**
10. **Confusing risk bands: 7.5%** (statin threshold with enhancers) vs **10%** (aspirin threshold) vs **20%** (high risk, statin).
11. **Time course of smoking benefit:** 50% $\downarrow$ within ~ 2 years, but **5-15 years** to reach non-smoker levels. It is not immediate.